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Project Report On

Multiple-Choice Biology Question and Distractor Generation Using T5 Transformer and Retrieval-Augmented Generation

Semester IX

DATA AND KNOWLEDGE MINING (IS-212) PROJECT

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CHAPTER 1

INTRODUCTION

1.1 Project Background

The rapid development of Artificial Intelligence (AI), especially Large Language Models (LLMs), has greatly changed how learning materials are created and the students are assessed. Among the various assessment methods, multiple-choice questions (MCQs) are widely regarded as the standard for evaluating students' comprehension of study material. A well-designed multiple-choice question with plausible distractors can encourage high-level thinking in students and help students remember information more effectively. However, manually creating high-quality multiple-choice questions and plausible distractors is a time- and resource-consuming process and requires expertise in the subject matter. In addition, low-quality distractors can be so obvious that they may fail to evaluate students effectively, which can lead to guessing instead of true understanding, and may affect the reliability and legitimacy of the outcome of the assessment.

This project explores the automated generation of multiple-choice Biology questions and distractors using Natural Language Processing (NLP) techniques. By combining strong capability of the Text-to-Text Transfer Transformer (T5) with Retrieval-Augmented Generation (RAG), this system automates both the creation of questions and the selection of plausible distractors. The goal of this project is to investigate the effects of different hyperparameter configurations when fine-tuning the T5 model for domain adaptation and semantic-preserving paraphrasing for diversity-enhanced question generation with a small corpus, to assess the feasibility of using the content of a single Biology text book to create a system capable of generalizing well enough to perform question and distractor generation for other books of the same domain, and to evaluate whether RAG provides an effective solution for distractor generation.

1.2 Objectives of the Project

The main objectives of this project are:

- To develop a T5 model capable of generating Biology questions from a given context paragraph and an answer keyword.

- To explore domain adaptation and diversity-enhanced, semantic-preserving question paraphrasing with minimal data by performing a second fine-tuning using a rehearsal technique to mitigate the effects of catastrophic forgetting.
- To implement a RAG system to retrieve semantically related but incorrect keywords to serve as distractors.
- To evaluate the system's performance in terms of question quality and distractor plausibility.

1.3 Scope of the Project

This project focuses on question generation, domain adaptation, and semantic-preserving paraphrasing within pre-university level Biology. The corpus used for second-stage domain-adaptation and paraphrasing fine-tuning is derived from a Cambridge International AS and A Level Biology coursebook. While the adaptation and paraphrasing data are restricted to Biology content, the T5 model is initially trained on the open-source SciQ dataset to obtain general question generation capability. SciQ contains questions from various scientific fields such as Physics and Chemistry. Therefore, the model is expected to generalize reasonably well across science-related subjects. However, its performance may degrade when applied to complex and content-dense contextual paragraphs from non-science domains, such as History or Business Management. Still, the model is expected to handle relatively simple and less complex paragraphs from other domains with reasonable performance.

On the other hand, the searchable database is constructed using the aforementioned Biology coursebook content to support the RAG-based distractor generation process. As a result, the effectiveness of the retrieval and distractor generation mechanism is currently limited to the pre-university level Biology domain. Nevertheless, the system can be extended to other subject areas by preprocessing additional domain-specific corpora and extending the existing searchable database.

Even though the primary question type considered in this project is multiple-choice questions, the system may also be adaptable to the generation of fill-in-the-blank questions.

1.4 Overview of the Methodology

This project is divided into two main parts – question generation and distractor generation. To support the question generation process, the T5 model was selected for its encoder-decoder architecture and unified framework that converts all text-based language problems into a text-to-text format. It is pretrained on the Colossal Clean Crawled Corpus (C4), a large-scale dataset of web text, to learn general language representations before task-specific fine-tuning [1]. Additionally, to overcome time and hardware constraints, Low Rank Adaptation (LoRA) fine-tuning was used, which significantly reduces the number of trainable parameters [2].

For question generation, the T5-Base model, with approximately 220 million parameters, was fine-tuned on the SciQ dataset to acquire question generation ability. To improve domain adaptation and paraphrasing quality, a second fine-tuning stage was conducted on a custom corpus using multiple hyperparameter settings to select the best-performing model, which balances between paraphrasing and retaining original meaning. The custom corpus was developed from the Cambridge International AS and A Level Biology coursebook content with the assistance of Gemini 3.0 Fast LLM. Additionally, a small set of SciQ samples was used in a rehearsal strategy to mitigate catastrophic forgetting during second fine-tuning stage.

For distractor generation, the coursebook content was first converted to markdown using the Docling library, then manually preprocessed, and finally split into serialized chunks with Docling. Each text chunk was embedded into a 300-dimensional word vector using the embeddinggemma-300m model. The resulting embeddings were indexed with Facebook AI Similarity Search (FAISS) library to create a searchable vector database, which was then used with maximal marginal relevance search to find the most relevant chunks for the MCQ. These relevant chunks were then fed into Mistral-Nemo-Instruct-2407, which is an instruction-tuned decoder-only generative transformer model, within RAG framework to produce plausible distractors. Subsequently, the similarity was calculated between the generated distractors and the actual answer of the MCQ to select the top three distractors which are incorrect yet semantically related to the answer.

1.5 Benefit of the Project

This project offers several significant values and insights to the educational and NLP landscape such as:

- **Time and Effort Efficiency:** Automates the labor-intensive process of drafting MCQs manually for educators.
- **Educational Support:** Improves learning outcomes by providing reliable and accurate assessment materials for both teachers and students.
- **Scalable Distractor Generation:** Uses RAG and vector database retrieval to efficiently produce semantically related distractors.
- **Resource Efficiency:** Shows that meaningful model performance improvements can be done by fine-tuning with a minimal dataset and the approach is feasible for low-resource educational domains.
- **Adaptability to Other Subjects:** The proposed framework can be extended to other academic subjects (especially science-related) by replacing the corpus.
- **Modular Framework:** The individual components (e.g., embedding model, retriever, or generative model) can be replaced or upgraded without the need to redesign the entire system.

CHAPTER 2

DATA PREPARATION AND PREPROCESSING

This chapter describes the datasets used in this project and the preprocessing steps carried out before fine-tuning and setting up the RAG pipeline. It presents statistical characteristics of the datasets, and explains how the data was prepared for question generation and RAG-based distractor generation.

2.1 Dataset and Data Statistics

This project utilizes two primary datasets for question generation and semantic-preserving paraphrasing – the SciQ dataset and a custom Biology corpus.

2.1.1 SciQ Dataset

The SciQ dataset is a publicly available collection of 13,679 crowdsourced multiple-choice exam questions. It covers topics in Physics, Chemistry, Biology and other scientific domains [4]. The dataset contains the following columns – question, correct_answer, distractor1, distractor2, distractor3, and support. The support column contains the context paragraph, and the correct_answer column contains the answer keyword for the generated question. However, the distractors from the dataset were not used in any of the fine-tuning in the project. Additionally, the SciQ dataset is intended for tasks such as question generation, distractor generation, reading comprehension, and question answering. The dataset is partitioned into 11,679 training samples, 1,000 validation samples, and 1,000 testing samples. To better understand the dataset, descriptive statistical analysis was performed, which resulted in important insights.

The Training Set contains 1,198 samples with missing context paragraphs, which were removed during the preprocessing step because they are unsuitable for question generation fine-tuning. Similarly, the Test Set and the Validation Set include 116 and 102 samples each, which were also removed during preprocessing. This means each of the data partitions had roughly 9-10% of the samples unusable. Luckily, none of the samples in any of the partitions had missing questions or answers.

The T5 Tokenizer was used to perform descriptive statistical analysis on the tokenized context paragraphs and questions. The following tables describe statistical measures including percentiles, mean, median, standard deviation, minimum, and maximum token counts.

Table I. Descriptive Statistics of SciQ Dataset Context Paragraph Token Counts

	Mean	**Median**	**Std**	**Min**	**Max**	**Q1**	**Q3**
Training Set	109.85	78	106.67	8	871	42	132
Validation Set	113.81	80	100.84	9	700	42.75	135
Test Set	108.01	80	109.22	9	622	41	134.5

Table II. Descriptive Statistics of SciQ Dataset Question Token Counts

	Mean	**Median**	**Std**	**Min**	**Max**	**Q1**	**Q3**
Training Set	17.44	16	7.46	4	106	12	21
Validation Set	17.91	16	7.86	6	70	13	21
Test Set	17.58	16	7.33	5	65	13	21

2.1.2 Biology Corpus

The Biology corpus was developed from the content of the Cambridge International AS and A Level Biology coursebook [3]. To generate the corpus, chapters and paragraphs suitable for creating multiple choice questions were first chosen. Subsequently, the paragraphs were given to the Gemini 3.0 Fast LLM one-by-one to generate samples. To produce samples with questions that balance clarity and paraphrasing, one-shot prompting technique was also used to guide Gemini understand the requirements. Prior to samples generation, Gemini was given a sample format, an example sample, and a set of constraints which included the following:

- ❖ The answer keywords must be short and concise, for example Photosynthesis, Osmosis, T Cells, and Transport Layer. Keywords with long phrases such as “Surface Area to Volume Ratio” must be avoided.
- ❖ Each question must be understandable independently of its context paragraph because students will not have access to paragraphs during the exam. For example, questions such as “Based on the flowering plants mentioned in the paragraph, …” must be avoided.

- In order to avoid excessive paraphrasing, the vocabulary used in questions must closely match that of the corresponding context paragraph. For example, replacing words such as divide and develop with proliferation and differentiation is not acceptable.
- Since the questions are intended to be MCQ format, they must be structured appropriately.

Similar to the samples in the SciQ dataset, each sample in the corpus contains a context paragraph, a correct answer, and a question generated from the context. Each sample also contains topic and subtopic of the context paragraph as metadata.

Example Sample

{"**context**": "Carbon monoxide combines irreversibly with haemoglobin, reducing the oxygen-carrying capacity of the blood. Nicotine stimulates the nervous system, increasing heart rate and blood pressure, and stimulating vasoconstriction, which reduces blood flow to the extremities.", "**answer**": "Haemoglobin", "**question**": "With which protein in the blood does carbon monoxide combine irreversibly?", "**topic**": "Animals", "**subtopic**": "Gas Exchange and Smoking"}

For each context paragraph, 4-6 samples were generated, and human subjective evaluation was used to choose the optimal samples that best align the aforementioned requirements. Subsequently, the selected samples were organized in chapters and stored in JSONL files. A total of 351 samples were created for the Training Set, and 100 samples were created for Test Set. Additionally, different chapters were used for the two sets. The number of samples generated per chapter varies due to chapter length, the number of suitable paragraphs, and the conceptual richness of the topic. In addition, chapters which focus a lot on practical experiments or procedural details were omitted since they are not suitable for generating multiple-choice questions. The following table shows which chapters from the coursebook were used for each set, the number of samples generated from each chapter, and under which topic and subtopic samples from each chapter belonged.

Table III. Distribution of Samples by Chapter, Set, Topic, and Subtopic

Chapter	Title	Set	Number of Samples	Topic	Subtopic
Chapter 1	Cell structure	Test Set	30	Cells	Cell Structure

Chapter 7	Transport in plants	Training Set	110	Plants	Transport
Chapter 8	Transport in mammals	Training Set	116	Animals	Transport
Chapter 9	Gas exchange and smoking	Training Set	65	Animals	Gas Exchange and Smoking
Chapter 10	Infectious diseases	Test Set	25	Animals	Diseases and Immunity
Chapter 11	Immunity	Test Set	25	Animals	Immune System
Chapter 13	Photosynthesis	Training Set	60	Plants	Photosynthesis
Chapter 18	Biodiversity, classification and conservation	Test Set	20	Plants and Animals	Biodiversity and Conservation

Once again, the T5 Tokenizer was used to perform descriptive statistical analysis on the tokenized context paragraphs and questions. The following tables describe statistical measures including percentiles, mean, median, standard deviation, minimum, and maximum token counts.

Table IV. Descriptive Statistics of Biology Corpus Context Paragraph Token Counts

	Mean	**Median**	**Std**	**Min**	**Max**	**Q1**	**Q3**
Training Set	99.75	94	35.58	27	192	74	123
Test Set	97.13	92	32.95	31	188	79	114

Table V. Descriptive Statistics of Biology Corpus Question Token Counts

	Mean	**Median**	**Std**	**Min**	**Max**	**Q1**	**Q3**
Training Set	24.36	25	4.64	10	37	21	27
Test Set	32.1	23	4.61	10	35	20	26

2.2 Data Preprocessing

In this project, the data preprocessing procedures for the SciQ dataset and the Biology corpus were lightweight. However, preparing the textual content of the entire biology coursebook for use in RAG involved heavy manual preprocessing.

2.2.1 Preprocessing of SciQ Dataset and Biology Corpus

The preprocessing procedure of the SciQ dataset involved handling samples with missing values, removing unused features, and reformatting each sample into the training sample required for T5 model.

Initially, samples with missing values in the ‘support’ column, which contains context paragraphs, were removed from SciQ since they are unusable for question generation fine-tuning. 1,198 samples were removed from the Training Set, and the Test Set and the Validation Set had 116 and 102 samples removed respectively. Subsequently, the three columns containing distractor values were removed because distractor generation is handled by the RAG and the distractors from the dataset were not used in any of the fine-tuning stages. Subsequently, each sample from the dataset was restructured into the following input-output format and stored in CSV format to be used in T5 fine-tuning.

Format Template

Input: generate a question: answer: [correct_answer] context: [support]

Output: [question]

Example Sample

Input: generate a question: answer: gene transcription context: Gene transcription is controlled by regulatory proteins that bind to regulatory elements on DNA. The proteins usually either activate or repress transcription.

Output: What is controlled by regulatory proteins that bind to regulatory elements on DNA?

The prefix ‘generate a question’ is used because T5 relies on task-specific prefixes to indicate the desired task during training and inference [1]. Additionally, the prefix ‘answer’ is placed before the prefix 'context' to ensure that the answer is included in case the context paragraph is long enough to get truncated during tokenization.

For the Biology corpus, the samples were also restructured into desired input-output format prior to training. In addition, the corpus required only minimal preprocessing such as removing noise from some of the paragraphs before feeding them to Gemini, refining some generated sample, and using human subjective evaluation to discard samples that did not meet with the requirements.

2.2.2 Preprocessing of the Biology coursebook for RAG

The preprocessing procedure for the Biology coursebook required tedious manual preprocessing and the use of Docling library. Docling is an open-source tool that provides a unified DoclingDocument data model for rich document representation and processing. It can parse common document formats and exports them to Markdown, JSON, and HTML by applying advanced AI for document understanding [5].

Initially, the coursebook was converted from PDF to a DoclingDocument using the Docling library and then exported to Markdown (MD) format. After that, the text data in markdown file was manually preprocessed to remove noise, correct formatting, and ensure consistency. Certain sections and content types were removed because they were irrelevant for MCQ generation or could interfere with embedding and retrieval. The following table summarizes the main types of content that were removed and the brief reason for their removal.

Table VI. Types of Content Removed During Preprocessing of the Coursebook

Content Type	**Description / Reason for Removal**
Page 1–8	Pages before the main content (book cover text, Table of Contents, how to use this book, and introduction, etc.) do not contain Biology content.
Page 512–End	Pages after the main content (appendices, glossary, index, recommended resources, answers to questions, etc.) as they noise without contribution to MCQ generation.

Artifacts	Artifacts such as footers, foot notes, headers, formula and image placeholders, and in-text table and figure references were removed because they fragment paragraphs and introduce noise.
Figure and Diagram Contents and Captions	Captions and callouts from diagrams and figures were removed because they can fragment paragraphs and introduce additional noise that might confuse the model.
Table Contents and Table Captions	When converted to Markdown, table contents often became fragmented, with table borders represented by the pipe symbol (|) and underscore (_) characters. These artifacts introduce noise to chunks. Some table captions and contents that contained sufficient information were transformed into paragraphs instead.
Sections	Several sections were removed. They include end-of-chapter questions, drawing instruction, learning outcomes, in-chapter questions, "you should be able to" statements, boxed experiments, and practical skills sections (e.g., constructing line graph, recording results, etc.).

After the long and tedious preprocessing, the Markdown file was turned into serialized chunks using the HybridChunker, which is the recommended chunker for use in RAG. In addition, chunks with less than 50 characters were discarded as an effort to further reduce noise. After that, embeddinggemma-300m, a text embedding model with approximately 300 million parameters, was used to embed each of the remaining chunks into a 300-dimensional word vector (embedding).

Embedding models map natural language to fixed-length vector representations which place semantically similar text near each other in the embedding space. They are primarily useful for semantic similarity, information retrieval, and clustering [6]. The embeddings were indexed using Facebook AI Similarity Search (FAISS) library to create a vector index (vector store). The vector index was stored in the local storage and used during RAG process to retrieve the most relevant chunks for the MCQ questions for distractor generation.

CHAPTER 3

APPLIED METHODOLOGIES AND MINING

This chapter explains the main methodologies and experimental procedures used in the project. It describes the two-stage fine-tuning process for question generation, and the RAG pipeline used for distractor generation and selection.

3.1 Experimental Setting

The experimental setting for this project involves multiple stages such as first-stage fine-tuning, second-stage fine-tuning, model selection, and retrieval setup. All of the experiments were conducted on Google Colab using an NVIDIA T4 GPU. In addition, mixed precision training (BF16) was employed.

Model performance was evaluated after each epoch using cross-entropy validation loss during the first fine-tuning stage. Although early-stopping was enabled, it did not affect training completion. On the other hand, a validation set was not used during the second-stage fine-tuning. This stage focused on paraphrasing and creative question generation, and standard evaluation metrics like validation loss or BLEU are not reliable indicators of output quality for creative tasks [7]. Additionally, the dataset for the second-stage fine-tuning was small and it could make validation metrics less meaningful and unstable.

Furthermore, during model evaluation, the baseline T5-Base model could not be used for comparison with fine-tuned models because it had not been fine-tuned for the task prefix and hence cannot produce any meaningful output. Consequently, all of the evaluations were conducted using the fine-tuned models.

3.2 Modeling

3.2.1 First-Stage Fine-Tuning

During the first-stage fine-tuning, only a single model configuration was trained on the SciQ dataset. Since the objective of this stage was to establish baseline question generation capability by enabling the model to understand the task-specific prefix, and because the validation loss showed stable convergence without signs of overfitting, extensive hyperparameter exploration was not performed. The Training Set with 1,2481 and the Validation Set with 898 samples, which were already formatted into the input-output format, were used. The following table presents the hyperparameter values that were used during this stage.

Table VII. Hyperparameter Values of the First-Stage Fine-Tuning Model

Hyperparameter	**Value**
Optimizer	AdamW (β_1=0.9, β_2=0.999, ε=1×10^{-8})
Learning Rate	3×10^{-4}
Label Smoothing	0.0
Effective Batch Size	64 (micro-batch size 8 × gradient accumulation 8)
Seed	42
Epochs	5
LR Scheduler Type	linear
Warmup Ratio	0.05
Weight Decay	0.01
LoRA Rank	16
LoRA Alpha	32
LoRA Dropout	0.1
Bias (LoRA)	none
Target Modules (LoRA)	q, k, v, o, wi, wo
Task Type	SEQ_2_SEQ_LM
Maximum Source Length	512
Maximum Target Length	256

Many of the questions in the SciQ dataset follows a relatively formulaic pattern, which constructs questions directly from the context sentence containing the answer keyword by removing the answer phrase and replacing it with ‘what’ and adding ‘?’ suffix. As a result, the model trained in this stage tended to generate structurally similar and somewhat crude questions. This limitation was addressed in the second fine-tuning stage, which focuses on paraphrasing.

3.2.2 Second-Stage Fine-Tuning

During the second-stage fine-tuning, the fine-tuned model obtained from the first-stage was further trained on the Biology corpus with three different hyperparameter configurations. Unlike the first-stage, this stage focused on acquiring a paraphrasing refinement and overcoming the aforementioned rigid question generation patterns.

The second-stage consisted of the Training Set of Biology corpus with 351 samples. In order to mitigate the risk of catastrophic forgetting and preserve the prior knowledge of scientific domains, a rehearsal strategy was applied. Catastrophic forgetting is a phenomenon in which a neural network loses the previously acquired knowledge when it is trained sequentially on new tasks or data. This is a common problem in sequential or continual learnings and strategies such as rehearsal, regularization, and parameter isolation have been proposed to mitigate it. Rehearsal, the strategy which was used in this stage, involves replaying the previous task samples while learning a new task to alleviate forgetting [8]. Therefore, 70 samples were chosen randomly from the preprocessed Training Set of the SciQ dataset and mixed with the Biology corpus, resulting in the Training Set with a total of 421 samples. Therefore, the replay samples made up of 20% of the second-stage Training Set.

A validation set was not used during this stage. This decision was made for two reasons. The first reason was that since this stage aimed to enhance paraphrasing diversity and creative question formulation, traditional metrics like cross-entropy validation loss and BLEU cannot reliably reflect output quality [7]. And the second reason was that due to the relatively small dataset, validation metrics could become unstable and less meaningful. Therefore, evaluation was conducted using a combination of metrics (BLEU, ChrF++, ROUGE-L, METEOR, and BERTScore), and LLM-based judgement to select the best-performing model. The table below presents the hyperparameter values of the models that were used during this stage.

Table VIII. Hyperparameter Values of the Second-Stage Fine-Tuning Models

Hyperparameter	**Model 1**	**Model 2**	**Model 3**
Optimizer	AdamW (β_1=0.9, β_2=0.999, $\varepsilon=1\times10^{-8}$)	AdamW (β_1=0.9, β_2=0.999, $\varepsilon=1\times10^{-8}$)	AdamW (β_1=0.9, β_2=0.999, $\varepsilon=1\times10^{-8}$)
Learning Rate	3×10^{-5}	3×10^{-5}	5×10^{-5}

Label Smoothing	0.0	0.0	0.0
Effective Batch Size	16 (micro-batch size 4 × gradient accumulation 4)	16 (micro-batch size 4 × gradient accumulation 4)	16 (micro-batch size 4 × gradient accumulation 4)
Seed	42	42	42
Epochs	5	10	5
LR Scheduler Type	linear	linear	cosine
Warmup Ratio	0.05	0.05	0.1
Weight Decay	0.1	0.1	0.05
LoRA Rank	32	32	32
LoRA Alpha	64	64	64
LoRA Dropout	0.1	0.1	0.05
Bias (LoRA)	none	none	none
Target Modules (LoRA)	q, k, v, o, wi, wo	q, k, v, o, wi, wo	q, k, v, o, wi, wo
Task Type	SEQ_2_SEQ_LM	SEQ_2_SEQ_LM	SEQ_2_SEQ_LM
Maximum Source Length	512	512	512
Maximum Target Length	256	256	256

3.3 Distractor Generation Using Retrieval-Augmented Generation

Distractor generation in this project uses Retrieval-Augmented Generation (RAG), which combines a retrieval mechanism with a generative LLM to produce distractors, which are semantically similar to the answer yet incorrect.

3.3.1 Vector Index

In order to set up the RAG system, the previously constructed FAISS-based vector index was loaded into the environment. The index, which contains the embeddings of the preprocessed Biology coursebook text chunks, was mapped to the same embedding model, embeddinggemma-

300m. This ensures that query vectors are generated in the same embedding space as the indexed vectors.

During distractor generation, the FAISS-based vector index performed semantic similarity search to retrieve text chunks relevant to the generated question. Max Marginal Relevance (MMR) search was used to balance relevance and diversity among retrieved chunks. The parameter ‘fetch_k’ of the ‘max_marginal_relevance_search’ function in the FAISS-based vector index was set to 20 to obtain the sufficient pool of semantically relevant candidate text chunks. The parameter ‘k’ was set to 4 and, the parameter ‘lambda_mult’ was set to 0.5 to retrieve four most relevant and diverse text chunks, which were then given to the generative LLM for distractor generation.

3.3.2 Generative LLM

The generative LLM of choice in this project is Mistral-Nemo-Instruct-2407, which is an instruct fine-tuned version of the Mistral-Nemo-Base-2407, with 12 billion parameters and it was trained jointly by NVIDIA and Mistral AI. According to its official model card, it significantly outperforms models of similar or smaller parameter size on multiple reasoning and QA benchmarks, showing both strong semantic generation capability and instruction following.

Due to hardware constraints and its large size, it was loaded using 4-bit NF4 quantization with double quantization. However, all of the computations were performed in bfloat16 for numerical stability. This approach made it possible to run the large model on a single NVIDIA T4 GPU without significant performance degradation. For prompting, the ChatML format, which was developed by OpenAI, was used because of its structured role-based format and compatibility with many instruction-tuned LLMs. The ChatML prompt given to the generative model was structured as follows:

```
<|im_start|>system
You are a biology teacher.
BASED ONLY ON THE PROVIDED CONTEXT, generate SIX distractors for
a multiple-choice question where the answer is '{target_word}'.
Rules:
1. Do not use synonyms of {target_word}.
2. Only use terms found in the context provided.
3. Format: List only the words with a newline at the end.
4. Do not put the numbers at the front.
<|im_end|>
<|im_start|>user
Context: {context_text}
Target Word: {target_word}
FIND SIX distractors:
<|im_end|>
<|im_start|>assistant
```

In the ChatML prompt above, the ‘target_word’ corresponds to the correct answer of the multiple-choice question, for which the model was tasked to generate distractors. The ‘context_text’ corresponds the semantically relevant chunks retrieved from the FAISS-based vector index.

The model outputs were generated using the Hugging Face generator function. The maximum number of output tokens was set to 125 to avoid overly long results. In addition, sampling was allowed for variation in outputs, and it was done by enabling the ‘do_sample’ parameter. Moreover, the ‘temperature’ parameter was set to 0.3 to reduce randomness and the parameter of Nucleus Sampling, ‘top_p’, was set to 0.9 to limit token selection to the most probable candidates while maintaining diversity. The ‘stop_sequence’ parameter was set to ‘<|im_end|>’ tag to ensure that generation terminates cleanly at the end of the assistant’s output.

3.3.3 Distractor Selection

After generation, the candidate distractors were preprocessed and compared against the correct answer using a similarity score. Distractors with similarity scores between 0.6 and 0.85 were considered the most suitable. In addition, distractors with scores below 0.55 were considered weak or irrelevant, while those with scores above 0.85 were considered too close to the answer and possible synonyms. However, if fewer than three suitable distractors were found, additional distractors were randomly sampled from those two groups to get a total of three distractors

CHAPTER 4

EVALUATION

This chapter presents the evaluation methods and results used to assess the performance of the proposed system. It includes evaluation metrics, human-based and LLM-based judgements, followed by the results and analysis on the closed test and open test.

4.1 Evaluation

In this project, the evaluation was performed only on the question generation component of the system. Evaluation metrics such as BLEU, ChrF++, ROUGE-L, METEOR, and BERTScore, along with LLM-based assessment of the generated questions were applied to measure the quality of the fine-tuned models. A more comprehensive evaluation of RAG-based distractor generation component is left for future work.

Additionally, two different test sets, a Closed Test Set and an Open Test Set, were used for evaluation. The Closed Test Set consisted of 100 samples developed from the Biology coursebook content. The Open Test Set consisted of a 300-sample subset randomly drawn from the SciQ Test Set. This set was used to evaluate the generalization capability of the model across other scientific domains. Due to time constraints, this project could not include a fully unseen benchmark dataset covering both scientific and non-scientific domains for a stricter out-of-domain evaluation.

For metric-based evaluation, all 100 samples from the Closed Test Set and all 300 samples from the Open Test Set were used. However, only 30 samples from each set were randomly selected for human-based assessment and LLM-based assessment due to its time-consuming nature.

4.2 Objective Evaluation

Objective evaluation involves measuring the similarity between generated questions and reference questions using metrics that calculates lexical overlap and semantic similarity between the generated and reference texts. Since question generation involves both semantic correctness and lexical variation (after paraphrasing), a combination of traditional and semantic-aware metrics were used. Although more advanced semantic metrics were used, traditional metrics such as BLEU and ChrF++ were also included for completeness and comparability with prior research.

4.2.1 BLEU

BLEU (Bilingual Evaluation Understudy) measures n-gram precision between the generated text and the reference text. It has been one of the most widely used evaluation metrics in NLP tasks. Although BLEU mainly focuses on surface-level token matching, it is included in this study to enable comparison with earlier question generation and text generation research that relied on this metric.

4.2.2 ChrF++

Unlike BLEU, ChrF++ computes a character level F-score using character n-grams, supplemented by word-level n-grams (typically up to 2-grams). Because it operates primarily at the character level, it is more tolerant of morphological variations and minor lexical changes. Although the primary purpose of including this metric was to enable comparison with earlier question generation studies that use this metric, it is still somewhat useful for evaluating paraphrased outputs.

4.2.3 METEOR

METEOR improves upon traditional n-gram-based metrics by aligning generated and reference sentences based not only on exact matches but also on semantically related words or tokens. Therefore, METEOR is more suitable for creative generation tasks where some degree of paraphrasing is expected.

4.2.4 ROUGE-L

ROUGE-L is a recall-oriented metric based on the Longest Common Subsequence (LCS). It measures how much of the reference text is captured in the generated text while preserving word order. In question generation, ROUGE-L helps assess if the generated question maintains the essential structure and key information of the reference question.

4.2.5 BERTScore

BERTScore evaluates similarity using contextual embeddings from pretrained transformer models. Instead of counting token overlap, it measures semantic similarity between tokens in embedding space. As a result, it can capture deeper semantic alignment between the generated and reference questions, and provides a more robust indicator of semantic correctness compared to traditional surface-level metrics.

4.3 Subjective Evaluation

In addition to objective metrics, subjective evaluation was also conducted since they do not provide enough insight into qualitative aspects of the generated questions such as grammatical fluency, contextual grounding, and precise alignment with the answer. In addition to the reference questions, both the context paragraph and the answer keyword must be taken into consideration when evaluating these aspects. Therefore, both LLM-based judgement and human-based evaluation were used to provide a more comprehensive assessment of question quality.

4.3.1 LLM-as-a-Judge

In order to complement the objective metrics, ChatGPT 5.2 Thinking, DeepSeek R1 DeepThink, and Gemini 3.0 Thinking were selected as three LLMs evaluators to assess the generated questions. Each LLM was given a prompt instructing it to evaluate questions based on the reference question, the context paragraph, and the answer keyword. The prompt asked the LLMs to score each generated question according to the following four criteria on a scale of 1 to 5:

- **Fluency:** measures grammatical correctness and naturalness of the question.
- **Context Relevance:** evaluates whether the question is grounded in the given context paragraph.
- **Answer Alignment:** checks whether the question clearly targets the given answer keyword.
- **Overall Quality:** a holistic judgement considering fluency, relevance, and answer alignment.

The generated questions from the 30-sample subsets of the test sets, along with their context paragraphs and answer keywords, were sent to the LLMs in 5-sample batches. Each question was independently evaluated by all three LLMs, and the scores of each criterion for each fine-tuned model were aggregated.

4.3.2 Human-Based Evaluation

To complement the evaluation metrics and LLM-based evaluation, human-based evaluation was also conducted. The same 30-sample subsets of generated questions, along with their corresponding context paragraphs and answer keywords from the Closed Test Set and Open Test Set, which were in LLM-based evaluation, were evaluated by three human assessors. Each assessor independently reviewed the generated questions using the same criteria applied in the LLM-based evaluation.

4.3.3 Inter-Rater Agreement (Fleiss's Kappa)

Since three evaluators were involved in each of the subjective evaluation processes, inter-rater agreement was used to measure the consistency of the evaluations. Fleiss's kappa was used to quantify the level of agreement among raters beyond chance [9]. Fleiss's kappa is commonly used when there are more than two evaluators, and a higher Fleiss's kappa value indicates stronger agreement among evaluators. Fleiss's kappa was calculated separately for each evaluation criterion. The Fleiss's kappa values can be interpreted by the following table:

Table IX. Interpretation of Fleiss's Kappa Values

Fleiss's Kappa Range	Interpretation
< 0.00	Poor agreement
0.00 – 0.20	Slight agreement
0.21 – 0.40	Fair agreement
0.41 – 0.60	Moderate agreement
0.61 – 0.80	Substantial agreement
0.81 – 1.00	Almost perfect agreement

4.4 Evaluation Results

4.4.1 Closed Test

Table X. Evaluation Metric Scores of Fine-Tuned Models on the Closed Test Set

	BLEU	ChrF++	METEOR	ROUGE-L	BERTScore
Stage-1 Baseline Model	**0.0856**	**6.4572**	0.2761	0.3510	0.8841
Model 1	0.0740	6.2049	0.3440	0.3805	0.8942
Model 2	0.0801	6.2060	0.3309	0.3953	0.8943
Model 3	0.0656	5.9020	**0.3642**	**0.4024**	**0.8980**

Table XI. LLM-Based Evaluation Scores (Average) of Fine-Tuned Models on the Closed Test Set

	Fluency	Context Relevance	Answer Alignment	Overall Quality
Stage-1 Baseline Model	**4.86**	4.73	**4.19**	**4.33**
Model 1	4.32	4.61	3.47	3.66
Model 2	4.77	4.34	3.6	3.78
Model 3	4.71	**4.76**	3.8	3.91

Table XII. Inter-Rater Agreement (Fleiss's Kappa) for the LLM-Based Evaluation on Closed Test Across Fluency, Context Relevance, Answer Alignment, and Overall Quality

	Fluency	Context Relevance	Answer Alignment	Overall Quality
Stage-1 Baseline Model	0.1883	0.3200	0.5128	0.3596
Model 1	0.4645	0.3128	0.5533	0.5655
Model 2	-0.0569	0.2861	0.5953	0.5527
Model 3	0.4114	0.0714	0.4850	0.5256

Table XIII. Human-Based Evaluation Scores (Average) of Fine-Tuned Models on the Closed Test Set

	Fluency	Context Relevance	Answer Alignment	Overall Quality
Stage-1 Baseline Model	4.533	4.089	3.944	3.911
Model 1	4.522	4.011	4.033	3.978
Model 2	**4.789**	4.122	4.111	4.111
Model 3	4.733	**4.567**	**4.656**	**4.589**

Table XIV. Inter-Rater Agreement (Fleiss's Kappa) for the Human-Based Evaluation on Closed Test Across Fluency, Context Relevance, Answer Alignment, and Overall Quality

	Fluency	Context Relevance	Answer Alignment	Overall Quality
Stage-1 Baseline Model	0.2969	-0.0968	-0.1222	0.0115
Model 1	0.0898	-0.0530	-0.0687	-0.1103
Model 2	-0.1330	-0.0223	-0.1429	-0.0774
Model 3	0.0672	-0.1867	-0.1264	-0.0626

4.4.1 Open Test

Table XV. Evaluation Metric Scores of Fine-Tuned Models on the Open Test Set

	BLEU	ChrF++	METEOR	ROUGE-L	BERTScore
Stage-1 Baseline Model	**0.2840**	**8.4681**	**0.5334**	**0.6021**	**0.9120**
Model 1	0.2091	7.5914	0.4692	0.3528	0.9025
Model 2	0.2043	7.1895	0.4846	0.4971	0.9022
Model 3	0.2089	6.9576	0.4751	0.4958	0.9025

Table XVI. LLM-Based Evaluation Scores (Average) of Fine-Tuned Models on the Open Test Set

	Fluency	Context Relevance	Answer Alignment	Overall Quality
Stage-1 Baseline Model	4.811	**4.922**	**4.667**	**4.578**
Model 1	**4.822**	4.522	3.922	4.022
Model 2	4.6	4.444	3.867	3.889
Model 3	4.3	4.6	3.822	3.833

Table XVII. Inter-Rater Agreement (Fleiss's Kappa) for the LLM-Based Evaluation on Open Test Across Fluency, Context Relevance, Answer Alignment, and Overall Quality

	Fluency	Context Relevance	Answer Alignment	Overall Quality
Stage-1 Baseline Model	0.2869	-0.0315	0.5647	0.3521
Model 1	0.2467	0.4351	0.5532	0.4706
Model 2	0.3065	0.3906	0.3704	0.4263
Model 3	0.2555	0.3895	0.3906	0.4870

Table XVIII. Human-Based Evaluation Scores (Average) of Fine-Tuned Models on the Open Test Set

	Fluency	Context Relevance	Answer Alignment	Overall Quality
Stage-1 Baseline Model	4.6	**4.478**	4.4	4.389
Model 1	4.7	4.267	4.233	4.333
Model 2	**4.722**	4.267	4.211	4.322
Model 3	4.511	**4.478**	**4.411**	**4.411**

Table XIX. Inter-Rater Agreement (Fleiss's Kappa) for the Human-Based Evaluation on Open Test Across Fluency, Context Relevance, Answer Alignment, and Overall Quality

	Fluency	Context Relevance	Answer Alignment	Overall Quality
Stage-1 Baseline Model	0.2139	-0.2734	-0.1917	-0.0719
Model 1	-0.1385	-0.1847	-0.1471	-0.1524
Model 2	-0.1037	-0.1255	-0.0736	-0.0478
Model 3	0.0862	0.0462	0.0986	0.0268

4.5 Evaluation Analysis

Only Model 1, Model 2, and Model 3 are compared to one another in Sections 4.5.1 and 4.5.2 because they are the second-stage fine-tuned variants developed under different hyperparameter settings. Since they belong to the same fine-tuning stage, comparing them separately makes it easier to identify which second-stage setting performs best. The First-Stage Fine-Tuned Model (Stage-1 Baseline Model) is discussed separately in Section 4.5.3.

4.5.1 Closed Test

In the closed test, the Model 2 obtained the highest BLEU (0.0801) and ChrF++ (6.2060), which means that there is higher lexical overlap with the reference questions. On the other hand, Model 3 achieved the highest METEOR (0.3642), ROUGE-L (0.4024) and BERTScore (0.8980), which shows better semantic similarity and contextual alignment. Because semantic metrics (METEOR, ROUGE-L, BERTScore) are more suitable for paraphrased question generation, Model 3 can be considered as the best-performing model on the closed test in terms of semantic quality even though it has lower lexical overlap.

As for LLM-based evaluation, Model 2 achieved the highest Fluency (4.77), while Model 3 achieved the highest Context Relevance (4.76), Answer Alignment (3.8), and Overall Quality (3.91). This shows that even though Model 2 generates questions with slightly better grammatical fluency, Model 3 produces questions that are much better in terms of contextual accuracy and alignment with the correct answers. Fleiss's kappa scores indicate fair-to-moderate inter-rater agreement across evaluation criteria, suggesting relatively consistent judgement from LLM evaluators.

The human-based evaluation supports Model 3 even more clearly. Although Model 2 achieved the highest Fluency score (4.789) again, Model 3 achieved the highest Context Relevance (4.567), Answer Alignment (4.656), and Overall Quality (4.589). This indicates that human raters generally preferred the questions produced by Model 3 in terms of their relevance correctness and overall usefulness for the closed test. However, the Fleiss's kappa values are very low and often close to zero or negative, which shows weak agreement among the human raters. Therefore, even though the human average scores favor Model 3, these findings should be interpreted with caution. Overall, when objective, LLM-based, and human-based results are considered together, Model 3 appears to be the strongest one for the closed test.

4.5.2 Open Test

In the open test, Model 1 achieved the strongest BLEU (0.2091) and ChrF++ (7.5914). It also achieved the highest BERTScore (0.9025) together with Model 3. Model 2 achieved the highest METEOR (0.4846) and ROUGE-L (0.4971). These results suggest that Model 1 retains stronger lexical similarity and stable semantic quality on other scientific domains. Model 2 performs slightly better in terms of meaning overlap and sequence-level similarity. Model 3, however, shows less consistent performance on other scientific domains.

For LLM-based evaluation, Model 1 achieved the highest scores in Fluency (4.822), Answer Alignment (3.922), and Overall Quality (4.022). This suggests that Model 1 also generalizes well for scientific domains other than biology. LLM-based Fleiss's kappa values are mostly fair to moderate, with relatively stronger agreement on answer alignment and overall quality, which supports the reliability of this assessment.

The human-based evaluation, on the other hand, presents a different view. Model 2 achieved the highest Fluency score (4.722), while Model 3 obtained the highest Context Relevance (4.478), Answer Alignment (4.411) and Overall Quality (4.411). Model 1 and Model 2 both scored slightly lower in Overall Quality, and Model 1 had the lowest average among the three models for most criteria in spite of its strong LLM-based evaluation performance. Additionally, Fleiss's kappa values are low overall, and Model 3 is the only model with positive agreement across all four criteria. However, low Fleiss's kappa values indicate weak inter-rater agreement. Therefore, the slight human-score advantage of Model 3 should be interpreted cautiously. Since objective and

LLM-based evaluations favor Model 1, it remains possible that Model 1 is the stronger model overall for the open test. As a result, the open test comparison is not conclusive.

4.5.3 Cross-Stage Performance Comparison

Interestingly, Stage-1 Baseline Model generally achieved stronger performance in overlap-based evaluation metrics and showed better robustness on the open test compared to the Second-Stage Fine-Tuned Models (Model 1, Model 2, and Model 3). This is primarily because its generation strategy closely follows the structure of the source sentence. This kind of behavior results in higher lexical overlap with reference questions, which improves overlap-based metrics such as BLEU, ROUGE-L, and ChrF++. In addition, Stage-1 Baseline Model also performed competitively in LLM-based and human-based evaluations, showing that its structurally consistent question formulation is clear, fluent, and contextually aligned. However, it exhibits limited paraphrasing ability and reduced linguistic diversity. In contrast, the Second-Stage Models introduce greater semantic variation and rephrasing, which may reduce lexical overlap but enhance expressive flexibility. Therefore, the conclusion can be done that paraphrasing increases diversity but does not necessarily guarantee higher evaluation scores.

CHAPTER 5

CONCLUSION

This chapter summarizes the outcomes of this project and presents its practical value and current limitations. It discusses the main advantages of the system, identifies its weaknesses, and suggests future work for its improvement.

Overall, this project demonstrated the development of an automated question and distractor generation system for pre-university Biology learning materials. By combining a two-stage fine-tuned T5 model with a RAG-based approach, the system was able to produce multiple-choice questions and distractor options with generally acceptable quality. Even though there are some limitations in consistency and generalization, the project still highlights the potential of AI-based educational content creation, which somewhat reduce manual effort.

5.1 Advantages of Using this Project

First, manually constructing MCQs that have plausible distractors is time-consuming, and requires subject matter expertise and considerable preparation time. By automating question generation with the T5 transformer and distractor generation with the RAG, this system provides an efficient tool for educators and students, especially in large-scale or resource-constrained environments.

Second, the understanding of the two-stage fine-tuning strategy was enhanced through a hands-on approach. The first-stage fine-tuning on the SciQ dataset enabled the model to obtain question generation ability, and the second-stage fine-tuning on the Biology corpus provided paraphrasing ability. The rehearsal strategy further helps preserve prior knowledge and mitigates catastrophic forgetting. This allows balanced domain adaptation and paraphrasing enhancement with minimal data.

Third, unlike generating distractors using only a language model's internal knowledge, the retrieval-based distractor generation ensures that distractors stay true to the context and reduces the risk of trivial or clearly incorrect options.

In addition, since the architecture of the system is modular, the components such as embedding model, vector index, generative LLM, and fine-tuned T5 model can be replaced or upgraded independently. This means that the system can be upgraded without a complete rework.

Moreover, the evaluation process is comprehensive. By combining evaluation metrics, LLM-based evaluation and inter-rater agreement analysis, it provides a balanced and multi-dimensional evaluation of question quality. This enhances the reliability of the experimental findings and provides deeper insights into question quality.

Finally, by replacing the domain corpus and reconstructing the vector index, the system can be extended or transferred to other science subjects and potentially to non-science domains.

5.2 Limitations

First, the RAG-based distractor generation relies heavily on a single Biology coursebook. As a result, the distractor plausibility is constrained by the coverage and depth of this coursebook. Performance may degrade when applied to different textbooks or advanced academic levels.

Second, the dataset used for second-stage fine-tuning is very small, and its reference questions were generated by Gemini 3.0 Fast LLM. This means that the diversity of linguistic patterns and question styles that the model can learn are limited.

Third, evaluating distractor generation and selecting the optimal distractors based on similarity score was not conducted extensively due to time and resource constraints. As a result, the effectiveness and plausibility of the generated distractors were not quantitatively validated.

In addition, although multiple LLMs were used and Fleiss's kappa was calculated to measure inter-rater agreement, the evaluation results may still not reflect human educator judgement.

Furthermore, hardware and computational constraints influenced architectural decisions. Generative LLM, Mistral-Nemo-Instruct-2407 in this case, had to be quantized and run under limited GPU memory. This may slightly affect generation quality. In addition, more extensive hyperparameter tuning experiments could not be done with the available computational resources.

Therefore, future work will focus on expanding the corpus, conducting comprehensive distractor evaluation, implementing question difficulty control, and evaluating the generated questions through educators.

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